Timestamp Analysis - Patient JC2106992

This file is one patient's record, pulled together from several linked tables that all reference the same three culture orders. It isn't a single clean table. It's more like six smaller tables stacked on top of each other, each keeping its own header row, all tying back to the patient's ID and to one of three order numbers. Every date in the file has been "jittered" for privacy, so the exact calendar dates have been nudged around a bit, but the spacing between events should still be roughly real.

The three culture orders

Everything in the file eventually points back to three culture orders, placed about a year apart in total. Here's how they line up:

Culture order	Date / time (UTC, jittered)	Gap from previous
Culture 1 (533739029)	2017-09-05, 21:59	(none, this is the first on record)
Culture 2 (570341032)	2018-08-15, 22:55	344 days (~11.3 months)
Culture 3 (576201136)	2018-10-04, 21:48	49 days (~1.6 months)

What stands out isn't so much the exact dates as the rhythm between them. There's almost a year of nothing between the first and second orders, then a much tighter six and a half week gap before the third. That's not an even, routine sampling pattern. It looks more like one isolated event, followed later by two that were closer together, possibly tied to the same episode.

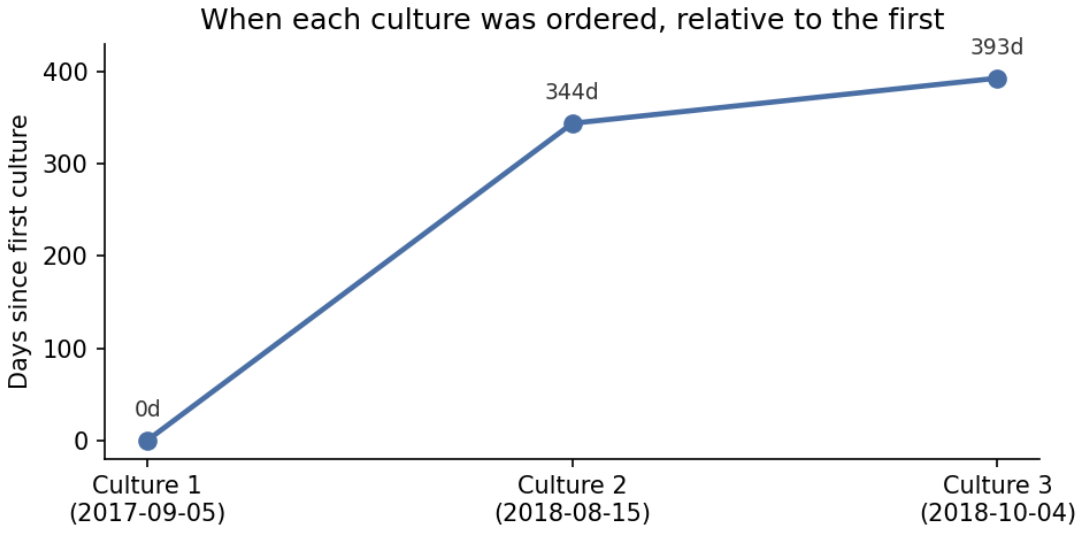


Figure 1: Each culture order, plotted as days since the first one.

How far back the medication history reaches

One of the more interesting fields is medication_time_to_culturetime, basically "how many days before this culture was ordered was this antibiotic given." It behaves differently across the three encounters, which is worth noting:

- Culture 1 only has a single, flat offset of 20 days for every medication attached to it, as if only one lookback window was checked at that point.
- Culture 2 has offsets of 29, 291, and 364 days, a much wider spread, some medications quite recent, others close to a year old.
- Culture 3 spreads out even further: 47, 79, 341, and 414 days back.

So the further along in the timeline you go, the richer the medication history that's actually available. The first culture seems to have had a narrow, fixed window checked, while the later two draw on a longer, more varied history.

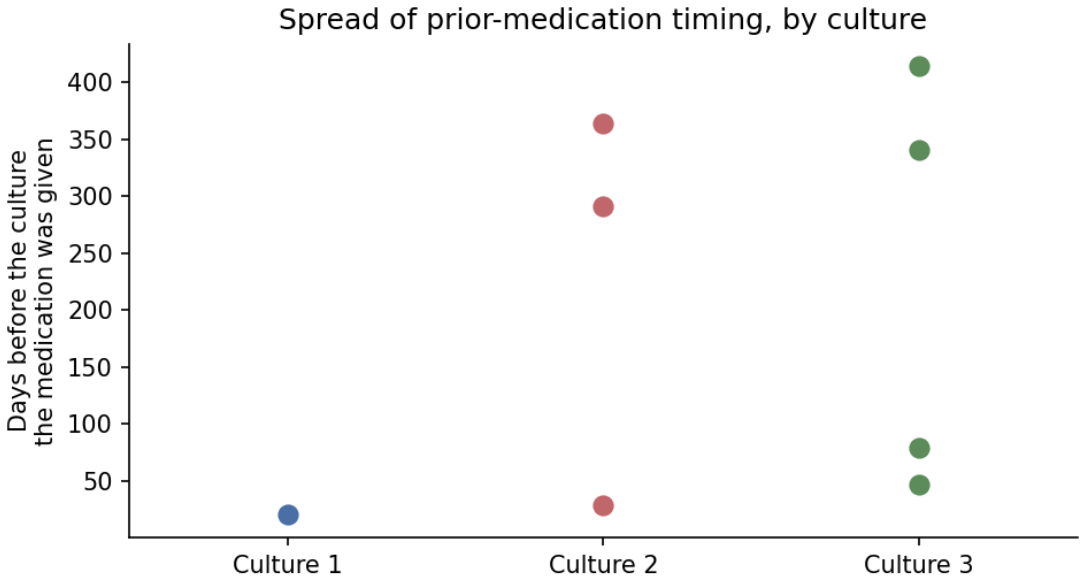


Figure 2: Where each culture's prior medications fall, in days before the order.

Everything else just reuses these three timestamps

The remaining tables, the ADI (deprivation index) scores, the antibiotic class and subtype exposure tables, and the larger susceptibility/cohort table, don't add new time information of their own. They each attach to one of the three existing culture timestamps as a kind of foreign key, mostly so the rows can be joined back to the right encounter. The cohort table in particular is the biggest block of rows in the file, about twenty organism/drug/susceptibility combinations, but time-wise it's flat. Every row just inherits whichever of the three dates its culture belongs to.

Overall picture

- One patient, three culture episodes, spanning a little over a year.
- Only two fields actually vary in time in a meaningful way, the medication lookback and the prior-organism lookback, and both are expressed as "days before the culture" rather than as calendar dates.
- The spacing between events is uneven: a long quiet stretch, then two events close together.
- The amount of history available grows over time. The first culture has a narrow, single-window lookback, while the later two pull from a much broader stretch of prior records.
- Nothing about the timestamps looks inconsistent or suspicious. The one cross-check available (the prior-organism offset) lines up with the actual first culture date almost exactly.